

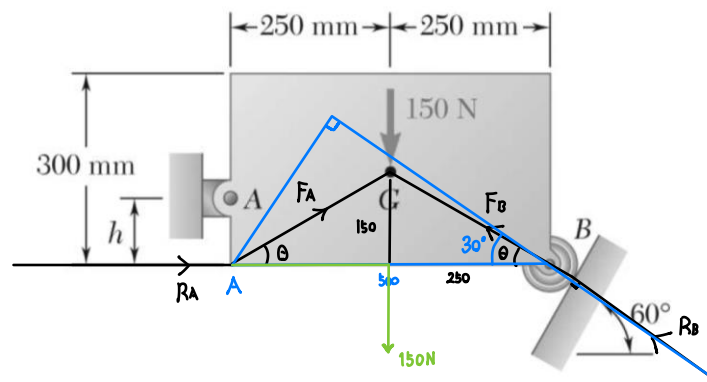
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Question 1

Determine the reactions at A and B in Figure 1 when

- i. $h = 0$ and [50%]
 ii. $h = 200$ mm. [50%]

**Figure 1**i. When $h = 0$,

$$F_{Bx} = F_{Ax} \quad \therefore F_B \times \cos \theta = F_A \times \cos \theta \quad \therefore F_B = F_A$$

$$F_{By} + F_{Ay} = F_B \times \sin \theta + F_A \times \sin \theta$$

$$= \frac{3\sqrt{34}}{34} (F_A + F_B) = 150 \text{ N}$$

$$\therefore F_A = F_B = 145.77 \text{ N}$$

$$\therefore R_A = \frac{F_A}{\cos \theta} = 170 \text{ N}$$

$$R_B = F_B \times \frac{\cos 30^\circ}{\cos \theta} = 147.22 \text{ N}$$

At point A

$$R_B \times (500 \text{ mm} \times \sin 30^\circ) = 150 \text{ N} \times 250 \text{ mm}$$

$$\therefore R_B = 150 \text{ N}$$

At point B

$$R_{Ay} \times 500 \text{ mm} = 150 \text{ N} \times 250 \text{ mm} \quad \therefore R_{Ay} = 75 \text{ N}$$

$$\therefore R_{Ax} = R_{Bx} = 150 \text{ N} \times \cos 30^\circ = 75\sqrt{3} \text{ N}$$

$$\therefore R_A = \sqrt{(75 \text{ N})^2 + (75\sqrt{3} \text{ N})^2} = 150 \text{ N}$$

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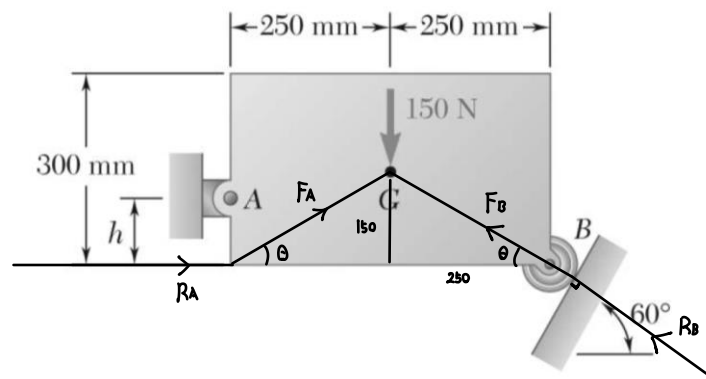
Question 1

Determine the reactions at A and B in Figure 1 when

- i. $h = 0$ and
- ii. $h = 200$ mm.

[50%]

[50%]

**Figure 1**i. When $h = 0$,

$$F_{Bx} = F_{Ax} \quad \therefore F_B \times \cos \theta = F_A \times \cos \theta \quad \therefore F_B = F_A$$

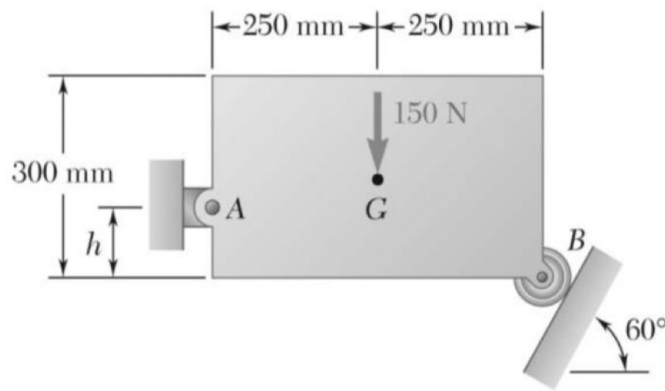
$$F_{By} + F_{Ay} = F_B \times \sin \theta + F_A \times \sin \theta$$

$$= \frac{3\sqrt{3}}{4} (F_A + F_B) = 150 \text{ N}$$

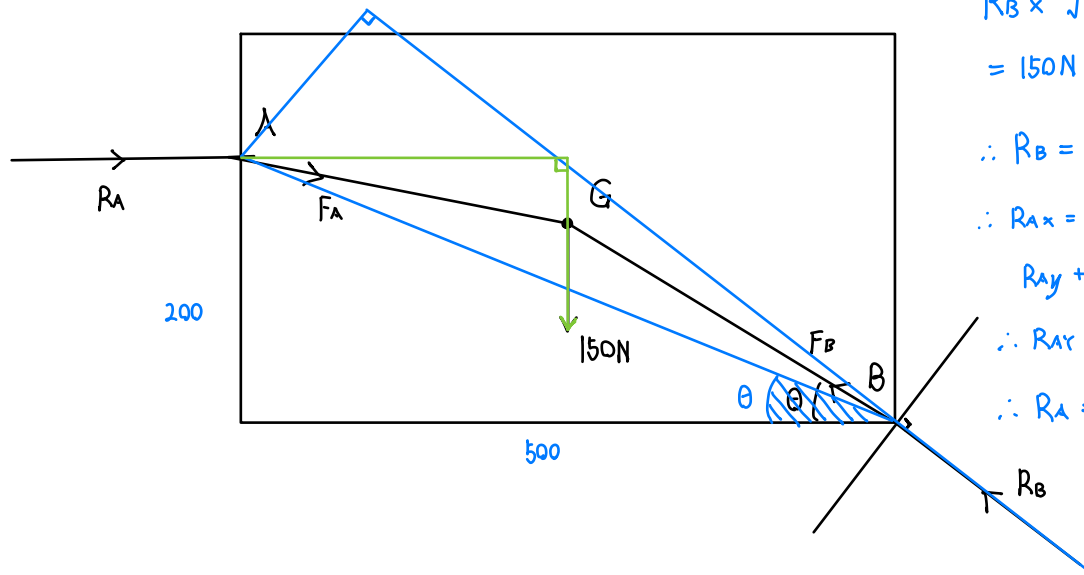
$$\therefore F_A = F_B = 145.77 \text{ N}$$

$$\therefore R_A = \frac{F_A}{\cos \theta} = 170 \text{ N}$$

$$R_B = F_B \times \frac{\cos 30^\circ}{\cos \theta} = 147.22 \text{ N}$$



ii When $h = 200 \text{ mm}$,



At point A

$$R_B \times \sqrt{500^2 + 200^2} \times \sin(30^\circ - \theta) = 150 \text{ N} \times 250$$

$$\therefore R_B = 488.31 \text{ N}$$

$$\therefore R_{Ax} = R_{Bx} = 422.89 \text{ N}$$

$$R_{Ay} + R_{By} = 150 \text{ N}$$

$$\therefore R_{Ax} = -94.16 \text{ N}$$

$$\therefore R_A = \sqrt{(R_{Ax})^2 + (R_{Ay})^2} = 433.25 \text{ N}$$

$$F_{By} = F_B \times \sin \theta = \frac{3\sqrt{34}}{34} F_B = 150 \text{ N} + F_{Ay} = 150 \text{ N} + \frac{50 \text{ mm}}{\sqrt{(250 \text{ mm})^2 + (50 \text{ mm})^2}} F_A = 150 \text{ N} + \frac{\sqrt{26}}{26} F_A$$

$$F_{Bx} = F_B \times \cos \theta = \frac{5\sqrt{34}}{34} F_B = F_{Ax} = \frac{250 \text{ mm}}{\sqrt{(250 \text{ mm})^2 + (50 \text{ mm})^2}} F_A = \frac{5}{26} \sqrt{26} F_A$$

$$\therefore 50\sqrt{34} \text{ N} + \frac{\sqrt{26}}{34} F_A = \frac{\sqrt{26}}{13} F_A$$

$$\therefore F_A = 75\sqrt{26} \text{ N} \quad \therefore F_B = 75\sqrt{34} \text{ N}$$

$$\therefore R_A = \frac{F_A}{\frac{250 \text{ mm}}{\sqrt{(250 \text{ mm})^2 + (50 \text{ mm})^2}}} = 390 \text{ N}$$

$$R_B = F_B \times \frac{\cos 30^\circ}{\cos \theta} = 441.67 \text{ N}$$